Akademia Górniczo-Hutnicza
im. Stanisława Staszica w Krakowie

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When Do Graph Neural Networks Help?

Class 3 — Literature & Experimental Protocol

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01 What we are building

Goal: answer one research question with a fair experiment — not "train as many models as possible".

Project in one sentence

When do graph connections help node classification, and when are simple tabular models (MLP / XGBoost) enough?

- **Task:** predict a node's class (e.g. paper topic)
- **Tabular models:** use node features only (ignore edges)
- **GNNs:** also use neighbours via edges
- **Stress tests:** remove real edges / add fake edges

02 Research question and hypotheses

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Research question

When do GNNs beat MLP / XGBoost, and how much edge noise can they survive before losing?

1. **H1 (breaking point):** on clean similar graphs GNNs win; after adding ∼30% random fake edges, MLP / XGBoost become better.
2. **H2 (similar vs mixed):** on mixed / heterophilous graphs, tabular models beat basic GNNs from the start.

03 Literature overview

Requirement: $\geq$3 papers / person, $\geq$6 unique papers in a pair.

- Anchor paper (Platonov et al., ICLR 2023) assigned to **M. Burda**
- Each student can explain claim, evidence, limitation, project use
- Full reading matrix: `LITERATURE.csv`

Michał Broś

- Huang et al. — C&S (ICLR'21)
- Kipf & Welling — GCN (ICLR'17)
- Veličković et al. — GAT (ICLR'18)

Michał Burda

- Platonov et al. — heterophily (ICLR'23)
- Hamilton et al. — GraphSAGE (NeurIPS'17)
- Zhu et al. — beyond homophily (NeurIPS'20)

04 Anchor paper



Platonov et al. (ICLR 2023)

A Critical Look at the Evaluation of GNNs under Heterophily

- **Claim:** popular heterophily benchmarks can be flawed (e.g. duplicates in Chameleon / Squirrel $\rightarrow$ leakage); on cleaner mixed graphs, strong simple models / standard GNNs often beat specialised heterophily GNNs.
- **Evidence:** duplicate analysis; new / revised datasets (incl. Roman-empire); MLP / GBDT-style vs GNN comparisons.
- **Limitation:** not a controlled edge-noise protocol.
- **Our use:** similar vs mixed setup; dataset choice; justification that tabular models can win.

05 Reading matrix (summary)

Assigned papers

Reader	Paper	Project use
M. Burda	Platonov ICLR'23	Anchor; homo/hetero; datasets
M. Bros	Huang C+S ICLR'21	Strong non-GNN comparator
M. Bros	Kipf GCN ICLR'17	Main GNN (Exp. A)
M. Bros	GAT ICLR'18	Attention GNN comparator
M. Burda	GraphSAGE NeurIPS'17	Main GNN (Exp. B)
M. Burda	Zhu NeurIPS'20	Why GNNs fail under heterophily

Full columns in CSV: claim and evidence, limitation, project use.

06 Datasets and splits

4 public graphs (PyG)

Dataset	Nodes	Same-class	Type
Cora	2.7k	81%	Similar
PubMed	19.7k	80%	Similar
Chameleon*	2.3k	23%	Mixed
Roman-empire	22.7k	5%	Mixed

* Prefer filtered Chameleon (Platonov) – no duplicates.

Splits

Fixed train / val / test
Same splits for all models
Preprocessing fit on train only

Resources

Laptop / Colab
$\sim$2–10 s per run
under 50 MB data

07 Models we compare

No graph (0% edges)

- MLP (features only)
- XGBoost / LightGBM
- Optional: MLP + Correct&Smooth

With graph

- GCN — averages neighbours
- GAT — learns edge weights
- GraphSAGE — samples neighbours

Start scope (MVP): Cora + one mixed dataset; MLP, XGBoost, GCN, GraphSAGE. GAT and C&S as extensions.

08 Metrics and fair protocol

- **Primary metric:** accuracy (node classification)
- **Secondary:** macro-F1 (if class imbalance)
- **Seeds:** 5 random seeds; report mean $\pm$ spread
- **Fair tuning:** same train/val/test; comparable HP budget
- **Logging:** settings, seeds, raw metrics, runtime / hardware

What would change our mind?

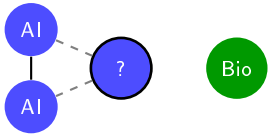
If GNNs stay ahead under high fake-edge noise on similar graphs, or beat tabular models on mixed graphs – H1 / H2 are weakened.

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09 Experiment A – deleting edges

Idea: randomly remove true edges: 0%, 20%, 40%, 60%, 80%, 100%.

- At 100% the GNN has no neighbours ≈ feature-only model
- Question: how much real structure is needed for GNNs to win?



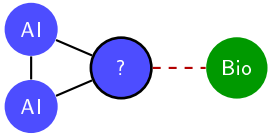
gray dashed = removed citations

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10 Experiment B – fake edges

Idea: keep true edges; add random wrong edges 0% $\rightarrow$ 75%.

- AI paper gets linked to a random Bio paper
- Hypothesis: after $\sim$30% noise, tabular models overtake GNNs
- Tabular baselines stay flat (they ignore edges)



red dashed = fake citation